Comparing Machine Learning Algorithms for ADHD Diagnosis Using Multimodal rs-fMRI and Clinical Features from the ADHD-200 Dataset

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***Abstract*— Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental condition marked by inattention and impulsivity, with increasing interest in noninvasive neuroimaging for diagnostic support. This study investigates the performance of several machine learning models in classifying ADHD versus control subjects using resting-state fMRI (rs-fMRI) data from the ADHD-200 dataset, comprising 223 individuals diagnosed with ADHD and 270 typically developing controls. Features were extracted from functional connectivity (FC) matrices (both full and PCA-reduced) as well as fractional amplitude of low-frequency fluctuations (fALFF) and an approximation of regional homogeneity (ReHo) calculated using Kendall's tau across 190 ROIs. Six models were evaluated: SVM (RBF kernel), Decision Tree, Random Forest, XGBoost, a multimodal deep neural network (DNN), and an early-fusion LSTM with attention. Each model was trained in combinations of imaging features and phenotypic data and evaluated using 10-fold cross-validation with a stratified 20% hold-out set. XGBoost achieved the highest performance with 94.89% accuracy and an F1-score of 0.9425, using PCA-reduced FC, ReHo, fALFF, and phenotype features. These findings highlight the value of integrating imaging and metadata and emphasize the importance of aligning feature types with model architectures for ADHD classification.**

Keywords*— *ADHD Classification, Resting-State fMRI, Functional Connectivity, fALFF, Regional Synchrony Approximation, Phenotypic Features, XGBoost, LSTM with Attention

I. Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental disorder affecting approximately 7% of the global population [1], often persisting from childhood into adolescence and adulthood. Core symptoms include persistent patterns of inattention, hyperactivity, and impulsivity, which can interfere with academic, occupational, and social functioning. Current diagnostic practices rely heavily on clinical interviews and behavioral assessments, which are subjective and may yield inconsistent results between clinicians. These limitations have spurred interest in developing objective, imaging-based biomarkers to support more consistent and reliable diagnoses.

Resting-state functional magnetic resonance imaging (rs-fMRI) has emerged as a promising tool for capturing intrinsic brain activity patterns in a task-free setting. In the context of ADHD, rs-fMRI studies have revealed alterations in large-scale networks such as the default mode network (DMN), frontoparietal control network, and salience network [2]. These findings have motivated the extraction of features such as functional connectivity (FC), fractional amplitude of low-frequency fluctuations (fALFF), and a regional synchrony measure approximating regional homogeneity (ReHo), computed using Kendall's tau correlations across ROI-level time series.

Machine learning (ML) techniques have shown considerable potential in leveraging these neuroimaging features to automate ADHD classification. However, the relative effectiveness of different ML models when applied to various rs-fMRI derived features remains underexplored. Moreover, the integration of subject-level metadata such as inattention scores, IQ measures, and medication status offers an opportunity to enhance model performance through multimodal learning.

In this study, we compare the classification performance of six machine learning models, including traditional classifiers and deep learning architectures, using a range of rs-fMRI based features and phenotypic data. By evaluating combinations of features and models on a well-curated subset of the ADHD-200 dataset, we aim to identify effective model-feature alignments and highlight design choices that improve generalizability and diagnostic utility.

II. Related work

The application of machine learning (ML) to resting-state fMRI (rs-fMRI) data has become increasingly common for the classification of Attention-Deficit/Hyperactivity Disorder (ADHD). A significant portion of prior work has focused on features derived from functional connectivity (FC), computed as Pearson correlations between region-level time series. These matrices, often extracted from atlases such as CC200 or AAL, result in thousands of features, typically reduced via principal component analysis (PCA) to mitigate overfitting and enhance training efficiency [3], [4]. Traditional ML models, including support vector machines (SVM) and Random Forests, have demonstrated consistent performance on these inputs. Some studies have also explored boosting-based methods such as XGBoost, though their use remains limited in ADHD-focused rs-fMRI research [3], [5].

Beyond FC, several studies have examined spectral and local synchrony features such as the fractional amplitude of low-frequency fluctuations (fALFF) and regional homogeneity (ReHo). These features are commonly extracted from 4D rs-fMRI volumes and reflect complementary information about localized and distributed brain activity. While ReHo is typically computed voxel-wise based on local temporal coherence, a few studies have introduced simplified, region-level approximations using Kendall's tau to reduce complexity and improve scalability [4], [6]. However, such ROI-level approximations remain underexplored and have not been systematically benchmarked against more established feature types.

Deep learning (DL) models, particularly convolutional neural networks (CNNs) and 3D CNNs, have been widely applied to fMRI classification tasks. These models are often trained on voxel-wise feature maps derived from ReHo or fALFF and have shown strong performance in some settings [7], [8]. However, their interpretability and generalizability remain limited, especially in multi-site studies with heterogeneous data. To address this, hybrid approaches combining CNNs with fully connected layers, transfer learning models such as ResNet, or attention mechanisms have been proposed, although these models rarely incorporate subject-level metadata during training [8].

A smaller number of studies have investigated multimodal learning strategies that integrate imaging features with phenotypic information such as IQ scores, ADHD subtype, or medication status. These models often apply feature-level fusion via concatenation or use metadata as separate input streams. While such approaches can improve classification accuracy, they typically treat imaging and non-imaging inputs independently rather than modeling interactions over time [3], [5], [9].

This study addresses these limitations by evaluating six classification models, including both traditional and deep learning methods, across combinations of FC, fALFF, ROI-level ReHo approximation, and phenotypic features. We further introduce an early-fusion LSTM with attention that incorporates metadata at each time step of the fMRI sequence, enabling the model to capture temporal dynamics in the context of individual clinical profiles. To our knowledge, this strategy has not yet been evaluated in ADHD-200–based classification studies.

III. METHODOLOGY

A. Dataset and Preprocessing

This study utilized data from the publicly available ADHD-200 dataset, which includes resting-state fMRI (rs-fMRI) scans and phenotypic information from multiple international sites. A total of 493 subjects were selected from sites that provided complete phenotype data, including both ADHD and typically developing control participants. These sites included Kennedy Krieger Institute (KKI), New York University (NYU), and three cohorts from Beijing University (Peking 1, Peking 2, and Peking 3). All imaging data had been preprocessed using the Athena pipeline [10], which performs slice timing correction, motion correction, nuisance signal regression, normalization to MNI152 space at 4 mm^3 resolution, and spatial smoothing with a 6 mm full-width-at-half-maximum Gaussian kernel. Bandpass filtering was applied in the range 0.009–0.08 Hz to isolate low-frequency BOLD fluctuations.

Region-of-interest (ROI) time series were extracted using the CC200 parcellation, which defines 190 functionally homogeneous brain regions. For each subject, the pipeline provided a time series matrix of varying length depending on scan duration. These raw time series were used directly for feature extraction steps such as functional connectivity (FC), fractional amplitude of low-frequency fluctuations (fALFF), and regional synchrony approximation. For LSTM-based models only, all-time series were zero-padded or truncated to a fixed length of 260 timepoints to enable sequence modeling.

B. Feature Extraction

Three types of features were extracted from the ROI-level time series provided by the Athena pipeline: functional connectivity (FC), fractional amplitude of low-frequency fluctuations (fALFF), and a regional synchrony approximation. These features capture complementary aspects of intrinsic brain activity respectively reflecting inter-regional coordination, spectral amplitude in low-frequency bands, and average temporal coherence of each region with the rest of the brain. All computations were based on the 190 ROIs defined by the CC200 parcellation.

1) Functional Connectivity (FC)

Functional Connectivity was computed using Pearson correlation coefficients between each pair of ROI time series, yielding a 190 × 190 symmetric matrix per subject. The upper triangular portion of this matrix, excluding the diagonal, was flattened in row-wise order, resulting in a 17,955-dimensional feature vector. For traditional ML models, dimensionality was reduced using principal component analysis (PCA), retaining enough components to explain 95% of the variance. For the multimodal deep learning model, the full 190 × 190 FC matrix was preserved as one input channel.

2) Fractional Amplitude of Low-Frequency Fluctuations (fALFF)

The Athena pipeline provided both unfiltered and bandpass-filtered time series (0.01–0.08 Hz). fALFF was computed per ROI as the ratio of the variance of the filtered signal to the variance of the unfiltered signal. This resulted in a 190-dimensional feature vector for each subject. For the multimodal deep learning model, these values were reshaped into a 190 × 190 matrix by repeating each value row-wise to match the input format used for all imaging channels.

3) Regional Synchrony Approximation:

To estimate inter-regional synchrony, Kendall's tau correlation was computed between each ROI's time series and all other ROIs. For each region, the mean correlation across all pairs (excluding self-comparison) was calculated. This produced a 190-dimensional vector representing each region's average synchrony with the rest of the brain. For the multimodal deep learning model, this vector was similarly reshaped into a 190 × 190 matrix to match the input shape used across all imaging feature channels.

All feature extractions were implemented in Python using NumPy and SciPy. Code is available at: [GitHub Repository].

C. Phenotypical Features

In addition to imaging-derived features, six subject-level phenotypic variables were included to provide demographic and clinical context: Inattentive, Hyper/Impulsive, Verbal IQ, Performance IQ, Full4 IQ, and Med Status. These features were selected from the phenotype data provided with the ADHD-200 dataset. Only sites with complete data for all six variables were included in the final sample, resulting in 493 subjects being used for training and evaluation. The five numerical features including Inattentive, Hyper/Impulsive, Verbal IQ, Performance IQ, and Full4 IQ; were standardized to zero mean and unit variance. The Med Status feature, which had two categories, was binary encoded. For traditional machine learning models, phenotypic features were concatenated with flattened imaging feature vectors to form a unified input. In the deep learning models, these features were handled in separate input branches: passed through a multi-layer perceptron (MLP) in the multimodal DNN and repeated across time steps in the early-fusion LSTM model to allow temporal alignment with the fMRI sequence.

D. Machine Learning Models

This study evaluated six classification models, including both traditional machine learning approaches and deep learning architectures. All models were trained using combinations of imaging features; functional connectivity (FC), fractional amplitude of low-frequency fluctuations (fALFF), and regional synchrony approximation, and standardized subject-level phenotypic features.

1) Traditional Machine Learning Models

Implementation of four traditional classifiers using scikit-learn and XGBoost: support vector machines (SVMs) with a radial basis function (RBF) kernel, decision trees, random forests, and XGBoost was done. Each model was trained and evaluated on the 493-subject subset of the ADHD-200 dataset using multiple structured feature sets. These included functional connectivity (FC), fractional amplitude of low-frequency fluctuations (fALFF), and regional synchrony approximation, both independently and in combination with subject-level phenotypic features. Phenotypic data were not used in isolation but were incorporated into fused input vectors to examine their additive impact on performance. For each subject, all features were concatenated into a single flattened vector, providing a standardized input format across models.

This evaluation allowed us to assess both algorithmic differences and the value of different feature combinations for ADHD classification. SVMs offered a strong non-linear baseline, while decision trees and random forests provided interpretable classifiers with built-in feature importance metrics. XGBoost, a gradient-boosted ensemble method, consistently achieved the highest classification accuracy across all feature configurations. It performed particularly well when trained on the full combination of imaging and phenotypic features, confirming its ability to model complex patterns and manage class imbalance in structured neuroimaging data.

2) Multimodal Deep Neural Network

The deep learning model consisted of two parallel branches: a 2D convolutional neural network (CNN) and a multi-layer perceptron (MLP). The CNN branch received a 3-channel 190 × 190 matrix input, representing FC, fALFF, and regional synchrony features. Each channel was processed through a sequence of convolutional layers with batch normalization, ReLU activations, and max pooling. A global average pooling layer was applied to extract spatial features, followed by an attention mechanism to weight feature importance. The parallel MLP branch received six standardized phenotypic features as a 1D vector input. This vector was passed through two dense layers with dropout regularization. The CNN and MLP outputs were concatenated and passed through fully connected layers with dropout, concluding in a sigmoid activation function for binary classification. The model architecture is illustrated in Fig. 1.

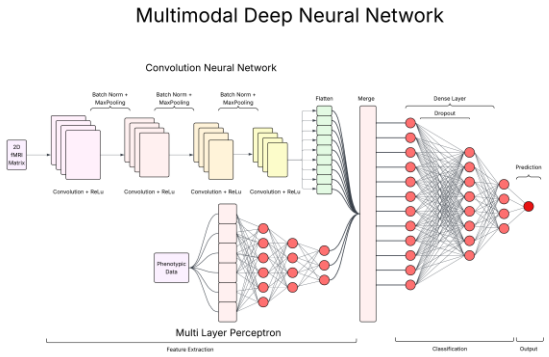


Fig. 1. Multimodal deep learning model combining 3-channel rs-fMRI features and phenotypic data via CNN and MLP branches with attention and late fusion.

3) Early-Fusion LSTM with Attention

To model temporal dynamics in resting-state fMRI signals, we implemented an early-fusion long short-term memory (LSTM) architecture with attention. The input consisted of 260 × 190 fMRI timepoints per subject, padded or truncated

as needed. The six phenotypic features were repeated across all time steps and concatenated with the fMRI signals, resulting in a 260 × 196 input matrix per subject. A masking layer was used to ignore padded time steps during training. The sequence was processed through two stacked LSTM layers with 64 and 32 units, followed by a dense attention mechanism that learned to weight informative time points. The resulting context vector was passed through fully connected layers with dropout and a final sigmoid activation for binary classification. A summary of all models, input features, and architectural characteristics is presented in Table I.

TABLE I. MODEL INPUTS AND ARCHITECTURAL OVERVIEW

Model Inputs and Architectural Overview		
Model	**Input Features**	**ML Architecture**
SVM (RBF)	PCA-reduced FC + fALFF + Regional Synchrony + Phenotypic Features	Kernel-based classifier using radial basis function
Decision Tree	PCA-reduced FC + fALFF + Regional Synchrony + Phenotypic Features	Single decision tree with Gini impurity-based splitting
Random Forest	PCA-reduced FC + fALFF + Regional Synchrony + Phenotypic Features	Ensemble of decision trees with majority voting
XGBoost	PCA-reduced FC + fALFF + Regional Synchrony + Phenotypic Features	Gradient-boosted decision tree ensemble
Multimodal DNN	3-channel rs-fMRI matrix (FC, fALFF, Regional Synchrony) + Phenotypic Features	CNN + MLP branches with attention and fusion layers
Early-Fusion LSTM	Padded fMRI Time Series + Phenotypic Features repeated across time (260 × 196)	Input-level fusion, stacked LSTM layers with attention and masking

E. Training and evaluation

This All models were trained and evaluated using a stratified 80/20 train-validation split, ensuring balanced representation of ADHD and control subjects. To reduce sampling bias and improve robustness, 10-fold cross-validation was performed during training, and evaluation metrics were averaged across folds. Model performance was reported using accuracy, precision, recall, and F1-score.

Hyperparameter tuning was conducted using randomized search across all models, including both traditional classifiers and deep learning architectures. Parameter selection was based on cross-validated performance within the training set, and the best configurations were selected for final evaluation. For deep models, early stopping was applied with a validation loss monitor to prevent overfitting, and dropout regularization was used in both CNN and MLP branches of the multimodal architecture.

Class imbalance was addressed during training by adjusting class weights. Traditional models were trained with weights derived from the distribution of diagnostic labels in the training data. For deep learning models, a fixed 2:1 weight ratio was applied, giving higher importance to the ADHD class. All deep networks were trained using the Adam optimizer with learning rates sampled from a predefined range. Following training, each model was evaluated on the validation set. Final model performance was evaluated on the 20% hold-out test set, and metrics including accuracy, precision, recall, and F1-score were computed using the best-trained configuration for each model.

IV. RESULTS

Table II presents the classification performance of all six models on the 20% hold-out test set, including accuracy, recall, precision, and F1-score. Each model was evaluated using its best-performing feature combination, as described in Table I. Among the traditional classifiers, XGBoost achieved the highest overall performance, reaching 94.89% accuracy and an F1-score of 0.9425. Random forest and decision tree models also performed strongly, achieving accuracies of 87.88% and 91.92%, respectively, with corresponding F1-scores of 0.8775 and 0.9194. These results highlight the strength of ensemble-based methods when applied to structured neuroimaging and phenotypic data and demonstrate that even relatively simple decision trees can yield competitive results when paired with well-engineered features.

In contrast, the deep learning models showed more modest performance. The early-fusion LSTM achieved an accuracy of 80.81% and an F1-score of 0.7765, outperforming the multimodal DNN, which reached 76.77% accuracy and an F1-score of 0.7554. While these results suggest that deep models can effectively incorporate temporal or multimodal structure, they also emphasize the continued effectiveness of classical ensemble methods for structured fMRI classification tasks.

TABLE II. CLASSIFICATION METRICS

Model	Accuracy	Recall	Precision	F1-Score
SVM (RBF)	74.57%	0.7475	0.7508	0.7426
Decision Tree	91.92%	0.9192	0.9227	0.9194
Random Forest	87.88%	0.8788	0.8829	0.8775
XGBoost	94.89%	0.9535	0.9318	0.9425
Multimodal DNN	76.77%	0.7677	0.7963	0.7554
Early-Fusion LSTM	80.81%	0.7500	0.8049	0.7765

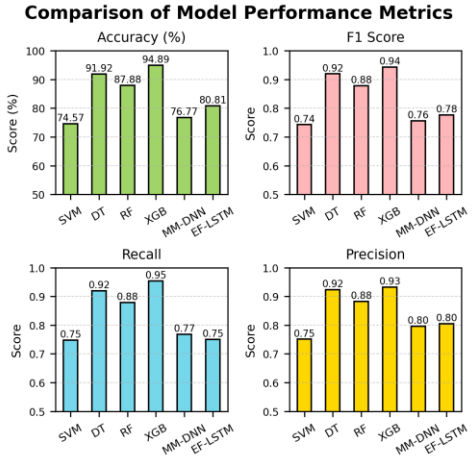


Fig 2. Classification metrics for each model on the 20% hold-out test set. Metrics include accuracy, recall, precision, and F1-score. Each model was evaluated using its best-performing feature combination from Table I.

V. Discussion

The results show that ensemble-based traditional models outperformed deep learning approaches in classifying ADHD from rs-fMRI and phenotypic features. This performance gap likely reflects the structured nature of the feature space, which favored tree-based algorithms that excel in handling mixed data types, non-linear interactions, and smaller training sets.

While the early-fusion LSTM offered modest improvements over the multimodal DNN, its performance remained below that of the top classical models. This may be due to the relatively low temporal resolution of region-level signals and the limited sample size available for training sequence-based models. Deep networks generally require larger datasets and more expressive input representations such as voxel-wise or graph-based formats; to fully capitalize on their capacity for learning temporal or spatial patterns.

These findings align with prior studies that have reported robust performance for XGBoost and Random Forest on functional connectivity features [3], [5] but extend them by demonstrating additional gains from combining FC with fALFF, a synchrony approximation, and phenotypic data. Notably, even simpler models such as decision trees achieved high accuracy when trained on well-engineered multimodal features, reinforcing the importance of feature selection over architectural complexity in data-limited settings.

Several limitations should be considered. The dataset was restricted to 493 subjects with complete phenotypic data, limiting both sample size and site diversity. No external validation set was used, and demographic factors such as age and sex were not explicitly modeled. These constraints affect generalizability and highlight the need for broader, more diverse datasets in future work.

Nonetheless, this study contributes a systematic comparison of classical and deep models using multimodal fMRI-derived inputs. The findings suggest that in scenarios where sample size is constrained and input features are structured, traditional machine learning remains a highly competitive option. Future work should explore voxel-level modeling, integrate domain adaptation to account for site differences, and evaluate the interpretability of both classical and attention-based models in greater depth.

VI. Conclusion

This study presented a comprehensive evaluation of traditional machine learning and deep learning models for classifying ADHD using resting-state fMRI and phenotypic features. The results demonstrated that ensemble-based classical classifiers, particularly XGBoost, outperformed deep learning models when trained on region-based imaging features derived from the CC200 atlas. These findings highlight the effectiveness of combining biologically meaningful fMRI metrics with subject-level metadata and reaffirm the strength of classical models in settings with limited data. Future work should explore voxel-level inputs, domain adaptation across imaging sites, and external validation to further improve model generalizability and clinical relevance.

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